serviceOrchestrationHistoryManagement

Service Description

Service ID: *"serviceOrchestrationHistoryManagement"*

Abstract
This document describes the `serviceOrchestrationHistoryManagement` service, version 1.0.0, produced by the DynamicServiceOrchestration core system of Arrowhead 5.2.2.

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1 Overview

This document describes the `serviceOrchestrationHistoryManagement` service, which provides query of the orchestration history: the pull and push orchestration jobs, their status and outcome. It is produced by the DynamicServiceOrchestration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2]. The MQTT binding is specified in the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

New in Arrowhead 5.

1.2 How This Service Is Meant to Be Used

Operators and LocalCloudManager inspect who requested which service when and with which result. In this workspace the job message names the selected providers (e.g. `1 local result: DataManager`), so the history shows who consumes what from whom.

1.3 Important Delimitations and Dependencies

Records older than `orchestration.history.max.age` days (default 15) are deleted.

2 Service Interface

This section describes the interfaces to the `serviceOrchestrationHistoryManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.



Figure 1: SysML-style block of the `serviceOrchestrationHistoryManagement` service and its operations.

The following interface operations are available.

2.1 operation query (OrchestrationHistoryQueryRequest) : OrchestrationHistoryResponse

Returns a page of orchestration jobs matching the filter.

3 Information Model

Here, all data objects that can be part of the `serviceOrchestrationHistoryManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.5, which are used to represent things like hashes and identifiers.

3.1 struct OrchestrationHistoryQueryRequest

Paged filter for orchestration jobs.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
ids	List<Identifier>	Job identifiers (UUID) to match.
statuses	List<String>	Job statuses to match.
type	String	Orchestration type to match: `PULL` or `PUSH`.
requesterSystems	List<Name>	Requester system names to match.
targetSystems	List<Name>	Target system names.
serviceDefinitions	List<Name>	Service definition names to match.
subscriptionIds	List<Identifier>	Subscription identifiers.

3.2 struct OrchestrationHistoryResponse

List of orchestration jobs.

Field	Type	Description
entries	List<OrchestrationJob>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.3 struct OrchestrationJob

One orchestration job (pull or push).

Field	Type	Description
id	Identifier	Identifier (UUID) of the orchestration job.
status	String	Status of the job: `PENDING`, `IN_PROGRESS`, `DONE` or `ERROR`.
type	String	Orchestration type: `PULL` or `PUSH`.
requesterSystem	Name	Name of the system that requested the orchestration.
targetSystem	Name	Name of the consumer system the orchestration was made for.
serviceDefinition	Name	Service definition name.
subscriptionId	Identifier	Subscription that started the push job.
message	String	Status message, e.g. `1 local result: DataManager`.
createdAt	DateTime	UTC time of creation.
startedAt	DateTime	UTC time when the job started.
finishedAt	DateTime	UTC time when the job finished.

3.4 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

3.5 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

4 References

[1] J. Delsing, “DynamicServiceOrchestration – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, serviceorchestration-dynamic/documentation/v5.2.2/DynamicServiceOrchestration-v5.2.2-SysD.pdf.

[2] ——, “serviceOrchestrationHistoryManagement HTTP/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, dynamicServiceOrchestration-v5.2.2-serviceOrchestrationHistoryManagement-HTTP-IDD.pdf.

[3] ——, “serviceOrchestrationHistoryManagement MQTT/TLS/JSON – Interface Design Description (DynamicServiceOrchestration),” Eclipse Arrowhead, IDD 5.2.2, 2026, dynamicServiceOrchestration-v5.2.2-serviceOrchestrationHistoryManagement-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	